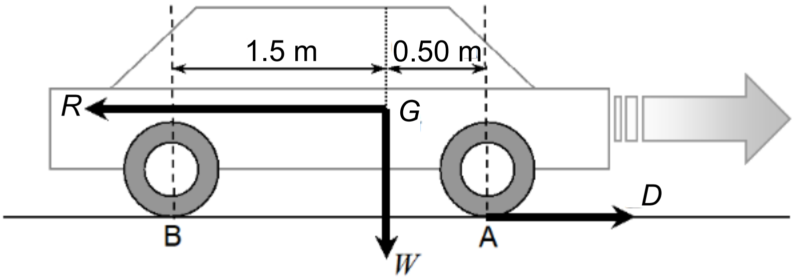
**10** The figure below represents the various forces acting on a car moving towards the right. The driving force, *D* acts on the front wheels and the total resistive force is represented by the force, *R*. The weight *W* of the car is 12000 N and it acts on the centre of mass, *G* which is 90 cm above the ground.




Given that the values of $D$ and $R$ are both 7000 N, what are the values of the normal reaction forces at A and at B acting on the wheels?

| | normal reaction force at A/ N | normal reaction force at B/ N |
|---|---|---|
| **A** | 8100 | 3900 |
| **B** | 6000 | 6000 |
| **C** | 6150 | 5850 |
| **D** | 5850 | 6150 |

Ans: D

Let $N_A$ and $N_B$ be the nornal contact forces at A and B respectively.

Taking moment about B,

$W(1.5) = R(0.90) + N_A(2.0)$

$$N_A = \frac{(12000)(1.5) - (7000)(0.90)}{2.0}$$

$= 5850$ N

Since vertical net force is zero, $N_A + N_B$=12000

$\therefore N_B = 12000 - 5850 = 6150$ N